Microsoft Elevate

CAPSTONE PROJECT

SALES ANALYSIS – AMAZON PRODUCTS

PRESENTED BY

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OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

With the large volume of products sold on the Amazon platform, it becomes difficult to monitor overall sales performance, identify high-performing product categories, and understand customer purchasing behavior. Without proper analysis, businesses may struggle to recognize seasonal sales trends and evaluate customer engagement through reviews.

PROPOSED SOLUTION:

- The proposed system aims to analyze and visualize Amazon product sales data to help stakeholders understand sales performance, product demand, and customer engagement. The solution leverages **data analytics and business intelligence tools** to generate meaningful insights and support data-driven decision making.

- The solution will consist of the following components:

- **Data Collection:**

- Gather historical sales data of Amazon products including **product category, product name, sales amount, number of products sold, reviews, and transaction dates**.

- Collect additional business attributes such as **month, week, quarter, and category information** to enable time-based and category-based analysis.

- **Data Preprocessing:**

- Clean and preprocess the dataset to handle **missing values, duplicate records, and inconsistent data formats**.

- Transform the data to create meaningful metrics such as:

- **YTD Sales (Year-to-Date Sales)**
- **QTD Sales (Quarter-to-Date Sales)**
- **YTD Products Sold**
- **YTD Reviews**

Perform **data modeling and relationships** to organize data efficiently for analysis.

Data Visualization & Analysis:

- Develop an interactive **Power BI dashboard** to present insights through visualizations.
- Implement key analytical visuals such as:

KPI Cards to display YTD Sales, QTD Sales, Products Sold, and Reviews.
Monthly Sales Trend to analyze sales performance across months.
Weekly Sales Trend to observe short-term sales patterns.
Sales by Product Category Table to compare category performance.
Top 5 Products by YTD Sales to identify best-selling products.
Top 5 Products by YTD Reviews to identify most popular products.

- **Provide interactive filters (slicers)** such as **Product Category and Quarter** to allow dynamic analysis.

Deployment:

- Deploy the dashboard on **Power BI Service** to enable easy access and sharing across stakeholders.
- Ensure the dashboard is **interactive, scalable, and user-friendly** so business users can quickly explore sales trends and make informed business decisions.

SYSTEM APPROACH:

1) MS OFFICE/EXCEL - VERSION 2021
2) POWER BI – VERSION 2026

ALGORITHM & DEPLOYMENT:

- The project uses **descriptive analytics and aggregation techniques within Power BI** to analyze sales data.
- Key calculations and measures are created using **DAX (Data Analysis Expressions)** to compute important business metrics such as:
- **YTD Sales (Year-To-Date Sales)**
- **QTD Sales (Quarter-To-Date Sales)**
- **Total Products Sold**
- **Total Reviews**
- These calculations help summarize large datasets and provide **clear insights into sales performance and product popularity**.

- **Data Input:**
- The input dataset contains Amazon product sales information, including:
- **Product Name**
- **Product Category**
- **Sales Amount**

Number of Products Sold
Customer Reviews
Date of Transaction
- Additional fields such as **Month, Week Number, and Quarter**
time-based analysis.

Processing & Analysis:
- The dataset is processed using **Power BI data modeling techniques**.
- Data transformation steps include:

Cleaning and formatting the dataset
Creating calculated columns and measures using **DAX**
Establishing relationships between tables
Aggregating data to compute **monthly, weekly, and yearly sales trends**
- These steps allow the dashboard to **efficiently analyze and summarize large volumes of sales data**.

Visualization & Prediction Insights:
- The processed data is visualized through **interactive dashboards** to highlight key insights.
- Important analytical visuals include:

KPI Cards showing overall sales metrics

Monthly Sales Trend Chart to identify seasonal patterns
Weekly Sales Chart to observe short-term performance changes
Sales by Product Category Table for category comparison
Top 5 Products by YTD Sales to identify best-selling products
Top 5 Products by Reviews to determine most popular products

- Interactive filters such as **Product Category and Quarter slicers** allow users to dynamically explore the data.

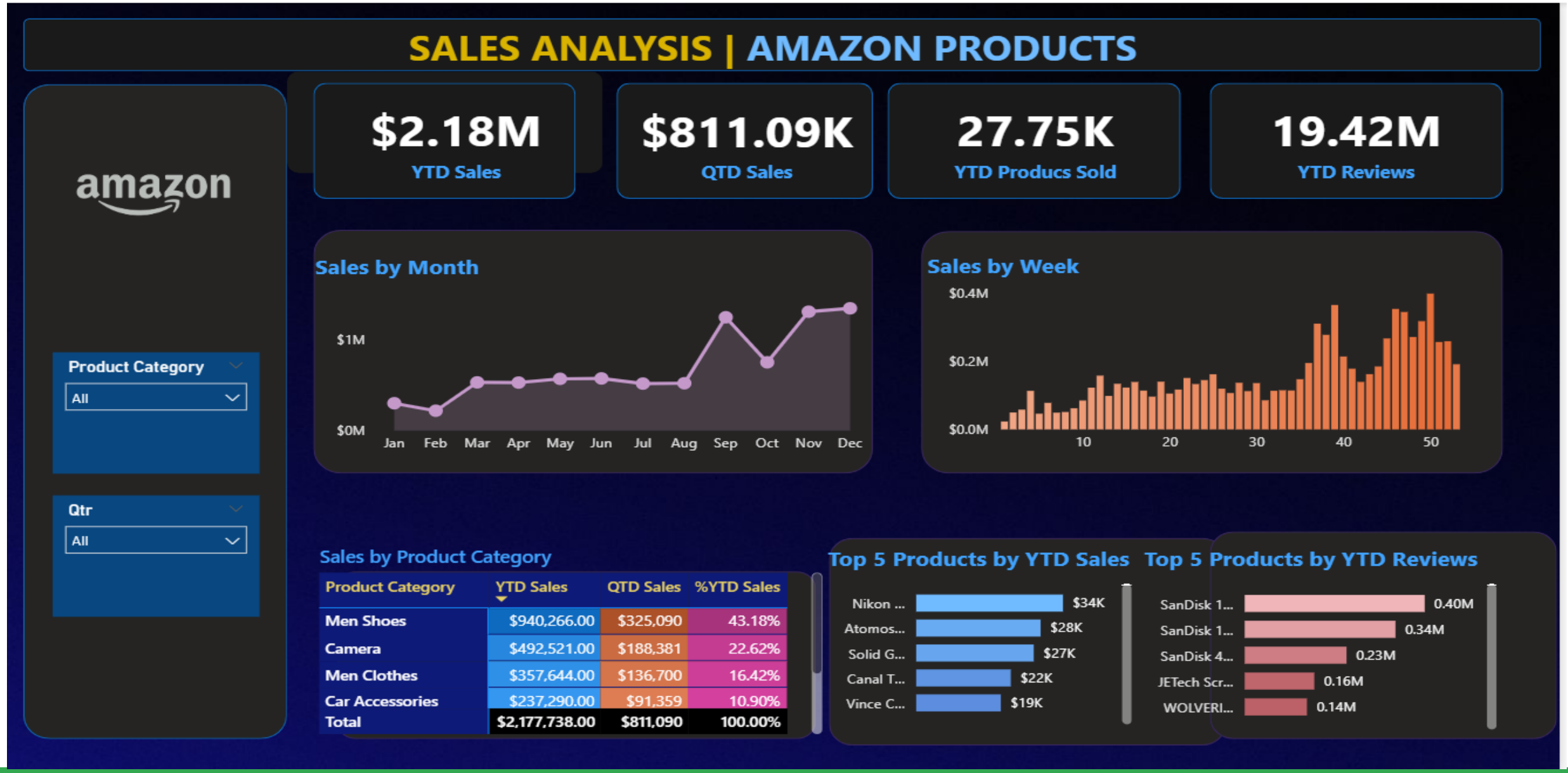
Deployment:

- The dashboard is deployed using **Power BI Service** so that stakeholders can easily access and share insights.
- Deployment steps include:

Publishing the dashboard from **Power BI Desktop to Power BI Service**
Enabling **interactive reports for business users**
Ensuring the dashboard is **scalable, secure, and accessible from web or mobile devices**

- This allows decision-makers to **monitor sales performance, analyze trends, and make data-driven business decisions efficiently**.

RESULT:


SALES ANALYSIS | AMAZON PRODUCTS
amazon
$2.18M
YTD Sales
$811.09K
QTD Sales
27.75K
YTD Producs Sold
19.42M
YTD Reviews
Product Category
All
Qtr
All
Sales by Month
$1M
$0M
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Sales by Week
$0.4M
$0.2M
$0.0M
10 20 30 40 50
Sales by Product Category
Product Category | YTD Sales | QTD Sales | %YTD Sales
Men Shoes | $940,266.00 | $325,090 | 43.18%
Camera | $492,521.00 | $188,381 | 22.62%
Men Clothes | $357,644.00 | $136,700 | 16.42%
Car Accessories | $237,290.00 | $91,359 | 10.90%
Total | $2,177,738.00 | $811,090 | 100.00%
Top 5 Products by YTD Sales
Nikon ... $34K
Atomos... $28K
Solid G... $27K
Canal T... $22K
Vince C... $19K
Top 5 Products by YTD Reviews
SanDisk 1... 0.40M
SanDisk 1... 0.34M
SanDisk 4... 0.23M
JETech Scr... 0.16M
WOLVERI... 0.14M

CONCLUSION:

- The **Amazon Sales Analysis Dashboard** provides a comprehensive view of product sales performance using interactive data visualizations.
- By analyzing key metrics such as **YTD Sales, QTD Sales, Products Sold, and Customer Reviews**, the dashboard helps in understanding overall business performance.
- The analysis identifies **top-performing product categories and products**, allowing businesses to focus on high-demand items.
- Time-based visualizations such as **monthly and weekly sales trends** help detect sales patterns and seasonal demand.
- Interactive filters enable users to **explore the data dynamically and gain deeper insights** based on category and time period.
- Overall, this dashboard supports **data-driven decision making**, helping businesses improve sales strategies, optimize product offerings, and monitor performance effectively.

FUTURE SCOPE:

- Integrate **real-time sales data** to monitor product performance and update insights automatically.
- Apply **machine learning models** to predict future sales trends and customer demand.
- Include additional data sources such as **customer demographics, pricing strategies, and promotional campaigns** to gain deeper business insights.
- Enhance the dashboard with **advanced analytics features** like customer segmentation and product recommendation analysis.
- Deploy the solution across **mobile and cloud platforms** to enable easy access for business users anytime and anywhere.
- Implement **automated alerts and notifications** to inform stakeholders about significant changes in sales performance.

REFERENCES:

- Amazon Sales Dataset used for this project - Dataset Link: Link

- Dashboard Presentation Power BI video - Video Link: Link

Thank You